

## Topics Covered (DSP Course)

### **Chapter 1:**

- Signals and systems
- Examples of digital signals
- Classification of Signals
  - Scalar and vector
  - Continuous and discrete
  - Analog and digital
  - Deterministic and random
- Typical signal processing operations
  - Scaling, delay, addition, integration, differentiation
- ~~Filtering~~
  - ~~Lowpass, high pass, bandpass, notch filter~~
- ~~Modulation~~
  - ~~Amplitude modulation~~
- Why DSP?
  - Advantages of DSP
  - Limitations of DSP
  - Application examples of DSP
- ~~Digital image signal~~

## **Chapter 2: Discrete-Time Signals in the Time Domain**

- Discrete-Time Signals- Time domain representation
  - Types of discrete-time signals
  - Size of a signal ( $L_p$ -Norm)
- Elementary operations on Sequence
  - Addition, multiplication, time-shifting, time-reversal (folding), branching, ensemble averaging,
  - Combination of basic operations
- Sampling rate alterations
  - Up-sampling, down-sampling
- Classification of sequences based on
  - Symmetry
  - Periodicity
  - Energy and Power
  - ~~○ Bounded and unbounded~~
  - ~~○ Absolutely summable~~
- Basic sequence
  - Unit impulse
  - Unit step
  - Real sinusoidal
  - ~~○ Exponential~~
  - ~~○ Two properties of sequences~~
- Sampling process
  - Ambiguity in sampling
  - Aliasing
  - Sampling theorem
- Correlation of Signals
- Properties of Autocorrelation and Cross-correlation Sequences
- Computation of Correlations
- ~~• Normalized Forms of Correlation~~
- ~~• Correlation computation for Power and Periodic Signals~~

## **Chapter 4: Discrete-Time Systems**

- Discrete-Time systems Examples
  - Accumulator
  - M-point moving average system
  - Exponentially weighted running average filter
- Linear interpolator
  - Factor of 2 interpolator
  - Factor of 3 interpolator
  - Median filter
- Classification of discrete-time systems
  - Linear and non-linear system
  - Shift-invariant and shift-variant system
  - Causal and non-causal system
  - Stable and unstable system
  - Passive and lossless system
- System Response
  - Impulse response
  - Step response
- Time-Domain characterization of LTI Discrete-time system
- Convolution sum
  - Linear method
  - Tabular method
- ~~• Stability Condition of an LTI Discrete-Time System~~
- ~~• Causality Condition of an LTI Discrete-Time System~~
- Simple interconnection scheme
  - Cascade connection
  - Parallel connection
- ~~• Finite-Dimensional LTI Discrete-Time Systems~~
  - ~~○ Total solution calculation~~
  - ~~○ Zero input response and zero state response~~
  - ~~○ Impulse response calculation~~
- ~~• Classification of LTI Discrete-Time Systems~~
  - ~~○ Based on Impulse response length~~
  - ~~○ FIR, IIR~~
- ~~• Based on the Output Calculation Process~~
  - ~~○ Non-recursive and Recursive System~~
- ~~• Based on the Coefficients~~
  - ~~○ Real Discrete-Time System~~
  - ~~○ Complex Discrete-Time System~~